

Benchmarking Foundation Models for Weather & Climate: Hands-on Session

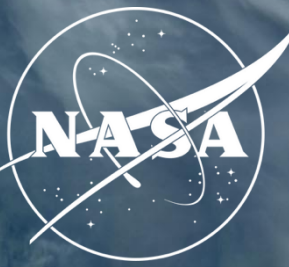
Katherine H. Breen
Global Modeling and Assimilation Office, NASA Goddard Space Flight Center
Morgan State University

Welcome and Orientation

Introductions

Session objectives and structure

Team Introduction



Science/Data Leads

- Lesley Ott (Integrative Modeling Virtual Institute PI)
- Mark Carroll (GSFC Data Science Team lead)



Support

- Alexander Kerr (Software Engineer)
- Michelle Frazier (Research/Data Scientist)

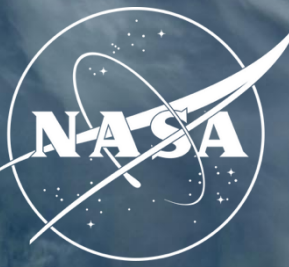
Hands-on Session (in person)

- Katherine H. Breen (Research Scientist)
- Jian Li (Research/Data Scientist)
- Melanie Frost (Data Scientist)

Special thanks to **Marshall Space Flight Center** for hosting and enabling this collaborative workshop.

We recognize the contributions of the **NASA Goddard Space Flight Center Data Science Group Innovation Lab** and **the Global Modeling and Assimilation Office (GMAO)** for their leadership in the co-development of benchmarking workflows for foundation models in weather forecasting, including rigorous evaluation against established standards and operational numerical weather prediction systems.

Orientation & Motivation



Foundation models (FMs) are rapidly advancing weather and Earth system prediction, with growing potential to complement/outperform traditional NWP.

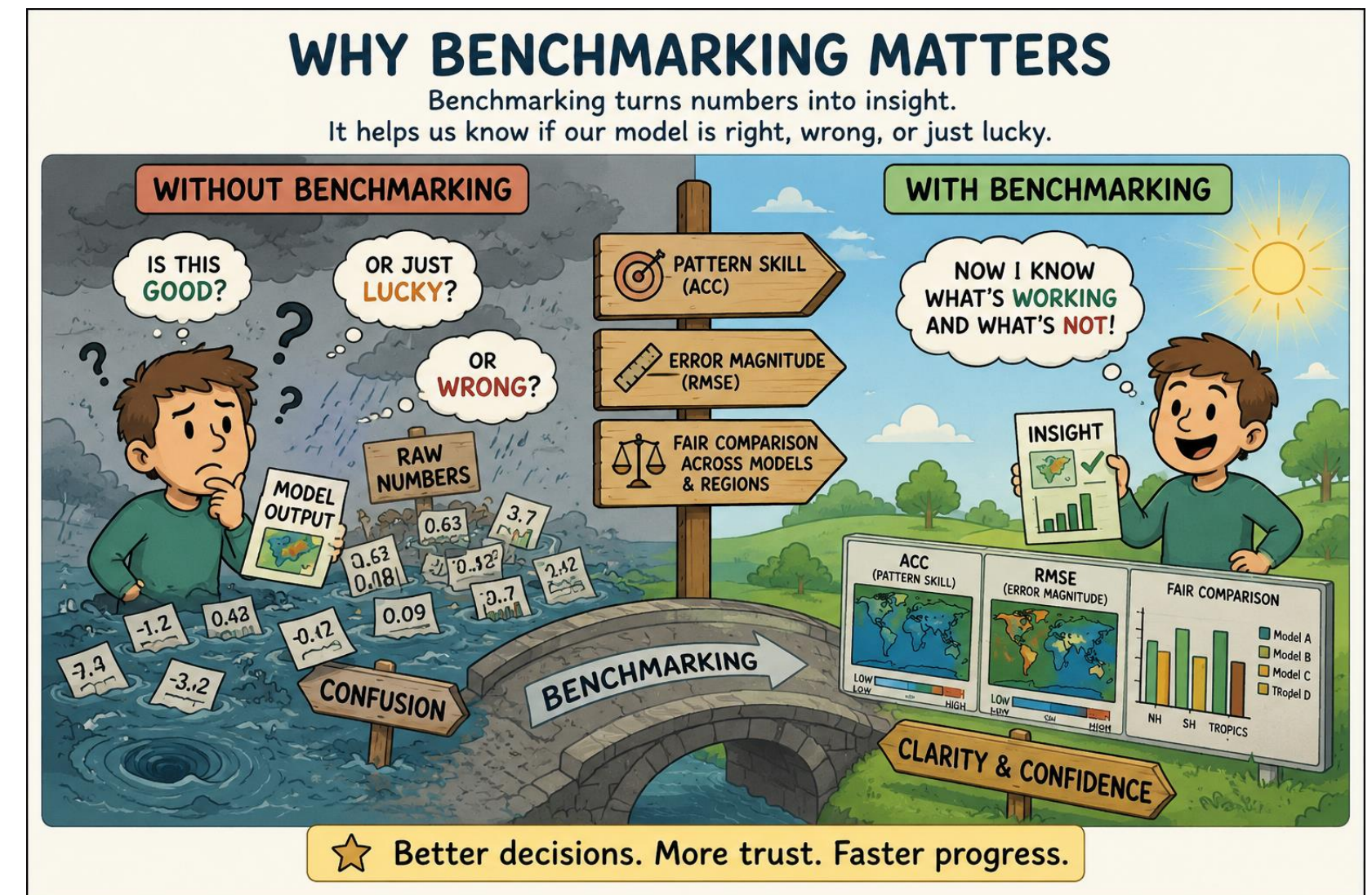
However, reported skill varies widely, making consistent evaluation critical.

Why benchmarking matters:

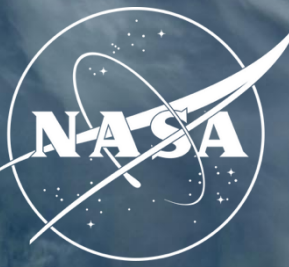
- Establish scientific credibility
- Identify strengths and limitations
- Assess readiness for research and operations

In this session, we will:

- Introduce foundation models for medium-range weather prediction
- Define key benchmarking approaches for meteorological forecasts
- Walk through the evaluation workflow used in this session



Session Goals



This hands-on session presents a science-driven framework for benchmarking state of the art (SOTA) foundation models for mid-range weather prediction.

Participants will:

- Generate 10-day forecast rollouts
- Evaluate FM forecast skill against reanalyses across pressure levels, regions, variables, and lead times
- Interpret forecast skill in a physically meaningful way
- Diagnose systematic biases and regime-dependent behavior

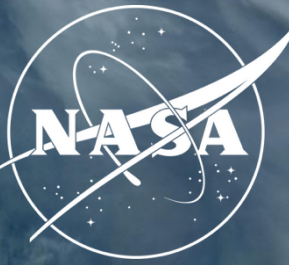
Outcome

Participants will gain practical tools and workflows for rigorous, science-based evaluation of foundation models in Earth system applications.

Key Takeaways

- Skillful forecasts are not enough without benchmarking
- Forecast skill depends on context
- Standardized workflows enable credible comparison

Session Agenda



8:00 AM: Welcome & Orientation

Introductions

Session objectives and structure

8:10 AM: Computing Environment Overview

Introduction to computational resources

Setup and workflow orientation

Begin spinning up instance for hands-on activity

8:20 AM: Background & Context

Weather prediction fundamentals

Role of foundation models in forecasting

8:30 AM: Hands-on Activity 1: Data Access & Forecast Generation

Reading reanalysis datasets (e.g., ERA5, MERRA-2)

Running medium-range foundation model rollouts

9:10 AM: Break (5 min)

9:15 AM: Hands-on Activity 2: Evaluation & Visualization

Using the statistics/metrics package

Generating diagnostic plots

9:45 AM: Background: Forecast Evaluation

Benchmarking approaches for weather and climate

Overview of forecast skill metrics

10:25 AM: Break (5 min)

10:30 AM: Discussion: Interpreting Forecast Skill

Global vs. regional performance

Vertical structure and regime-dependent behavior

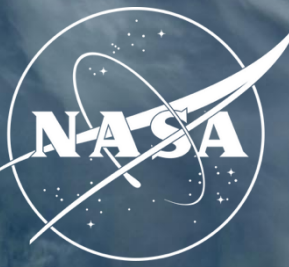
11:00 AM – 12:00 PM — Open Work Session

Individual coding time

1:1 support

Small group discussions

Next steps



Introduction to Sagemaker and JupyterLab computing resources

[10 min]

- Demo how to connect to computing environment
- Participants verify functionality
- Overview of directory structure, files provided, etc.

Reading and preparing input datasets

[10 min]

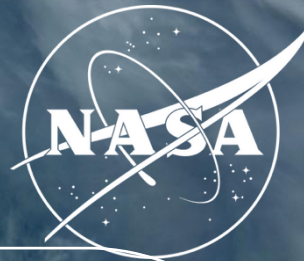
- Demo how to read data on-the-fly from CDO and how to generate a request, which collections to use, point to documentation wiki for ERA5, etc.
- For this session, we will read in *.nc files that we have staged ahead of time.

Background and Context

Atmospheric dynamics and weather prediction fundamentals

Role of foundation models in forecasting

Spatiotemporal Modeling Scales



Types of Forecasts

Weather Forecast

Seasonal to Interannual Prediction

Decadal: Climate Nowcast to Projection (Scenario-based)

Key model components



Atmosphere



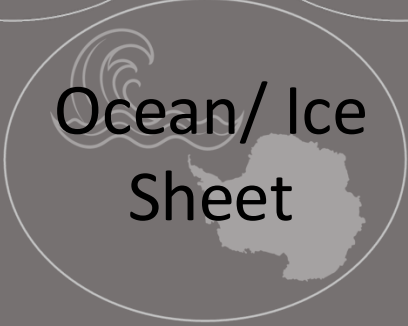
Land -
Hydrology



Atmosphere



Land/
Vegetation



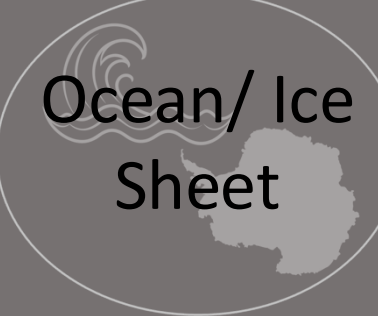
Ocean/ Ice
Sheet



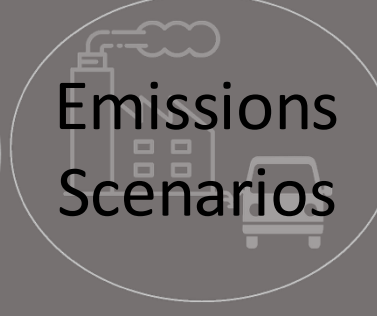
Atmosphere



Land/
Vegetation

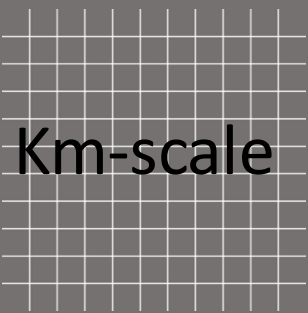


Ocean/ Ice
Sheet



Emissions
Scenarios

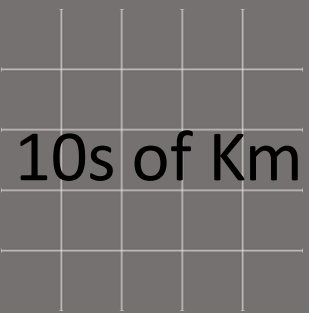
Spatial Resolution/ Update Frequency



Km-scale



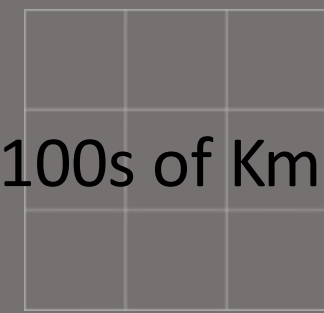
Hours



10s of Km



Monthly



100s of Km



?? – 5+ years

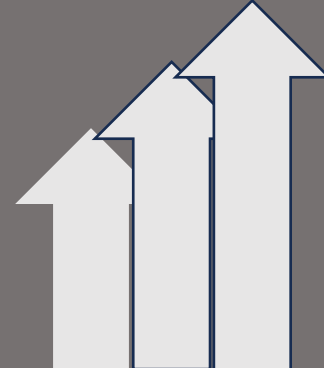
Ensemble Size/ Uncertainty Quantification



Initial
Condition

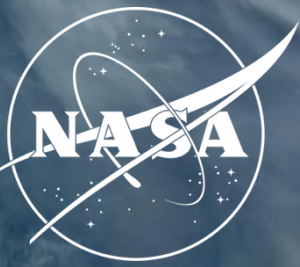


Initial/
Parametric



Parametric/
Structural

1922 Vision of Environmental Prediction

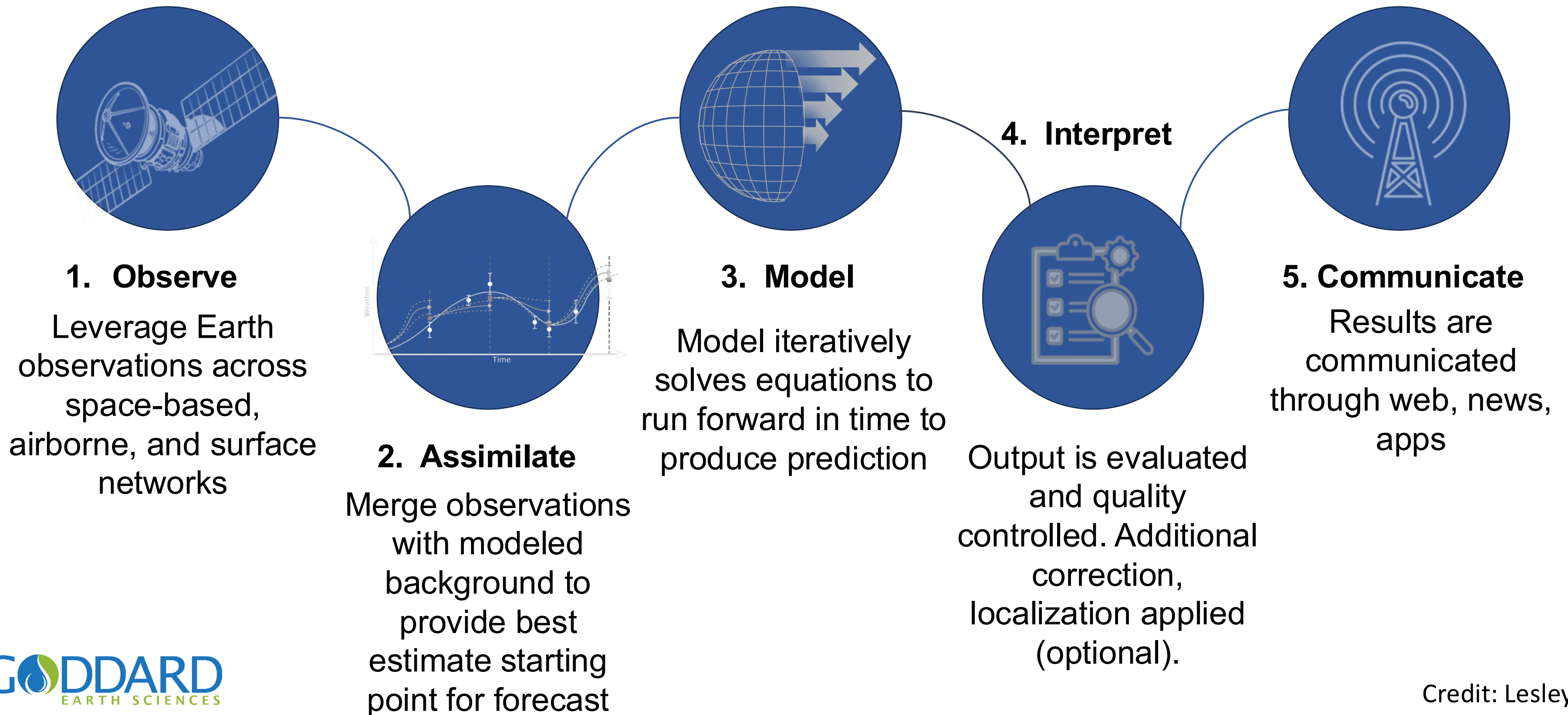
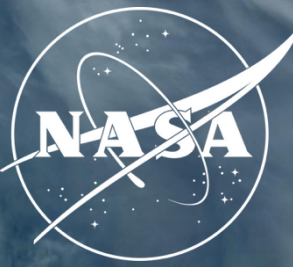


Lewis Fry Richardson's Human Powered Forecast Factory

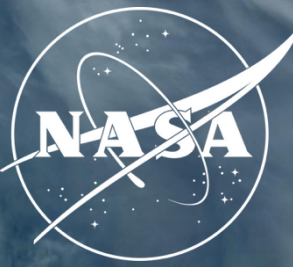
Powered by 64,000 human computers continuously solving equations of motion using data from surface stations, weather balloons



Forecast Factory in the Modern Era: Powered by data, computational infrastructure, and SOTA models



Timeline of Weather Forecast Evolution



1960s-1970s: advent of Numerical Weather Prediction (NWP)

First calculation of NWP on a computer: Charney, J. G., Fjoertoft, R. & Neumann, J. v. Numerical integration of the barotropic vorticity equation. *Tellus* 2, 237–254 (1950).

1980s: Use of statistical methods and improved numerical schemes in weather forecasting

Robert, A. J. A semi-Lagrangian and semi-implicit numerical integration scheme for the primitive meteorological equations. *J. Meteorol. Soc. Jpn* 60, 319–324 (1982).

1990s: Emergence of neural networks, sophisticated data assimilation

Daley, R. *Atmospheric Data Analysis* (Cambridge Univ. Press, 1991).

Rumelhart, D. E., Hinton, G. E. & Williams, R. J. Learning representations by back-propagating errors. *Nature* 323, 533–536 (1986).

2000s: ML enhancements to data assimilation and advanced data assimilation algorithms

Grell, Georg A., and Dezső Dévényi. "A generalized approach to parameterizing convection combining ensemble and data assimilation techniques." *Geophysical Research Letters* 29.14 (2002): 38-1.

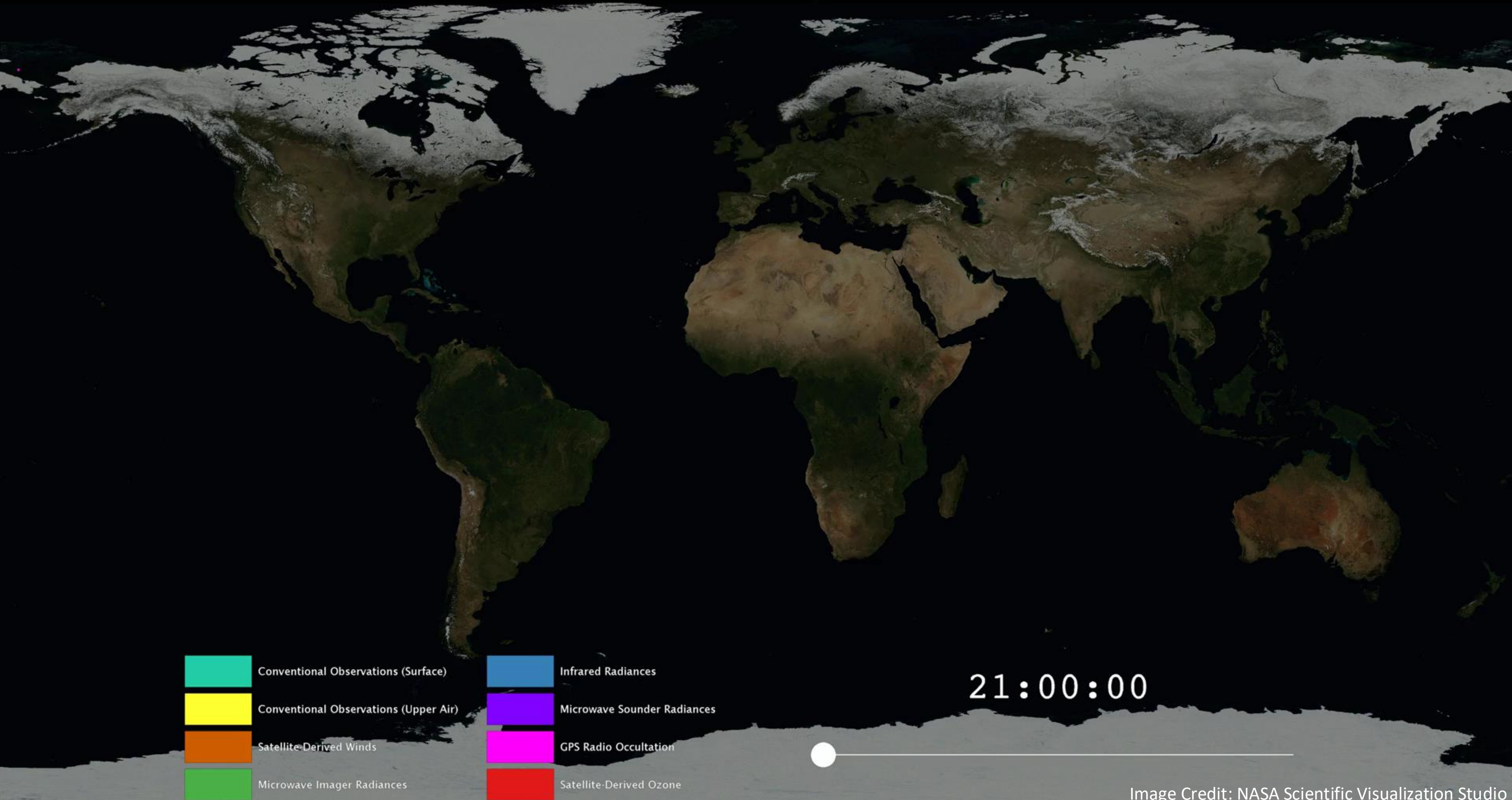
2010s: Deep learning applications in weather forecasting, hybridization of NWP and ML models

O'Gorman, Paul A., and John G. Dwyer. "Using machine learning to parameterize moist convection: Potential for modeling of climate, climate change, and extreme events." *Journal of Advances in Modeling Earth Systems* 10.10 (2018): 2548-2563.

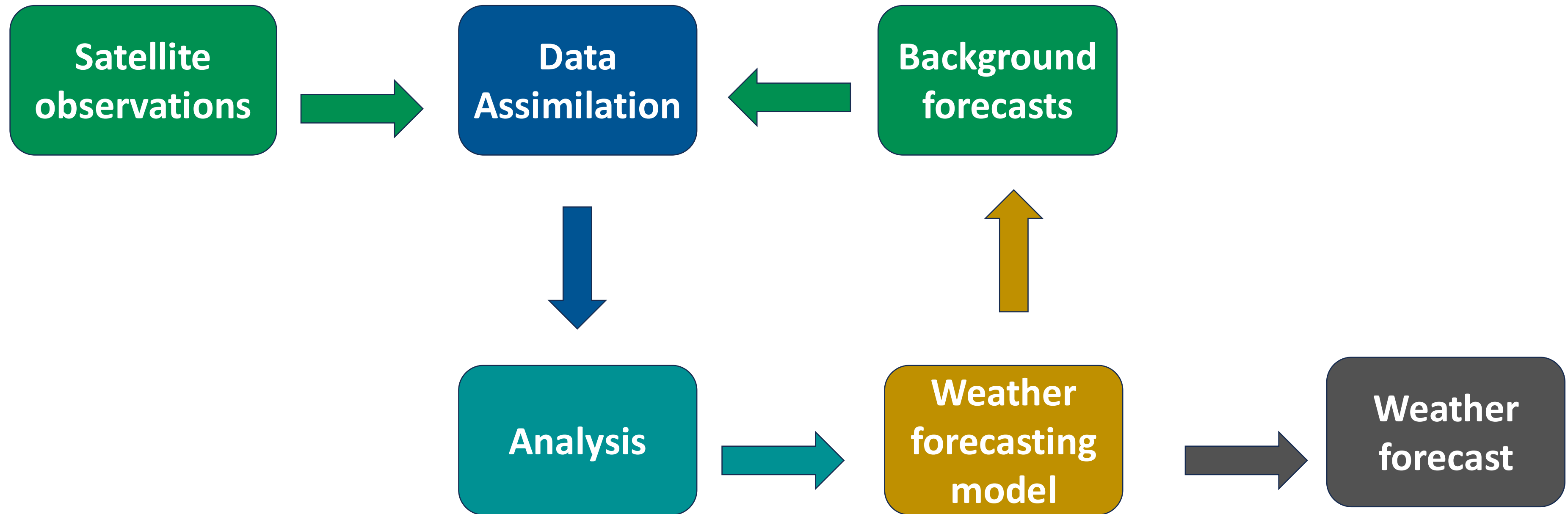
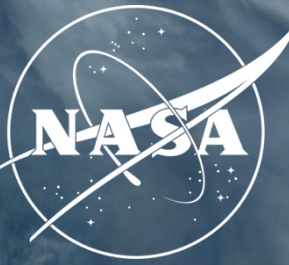
2020s: Real-time forecasting and generative models, rise of explainable AI, early development of Earth System Digital Twins (ESDTs) and Foundation Models for weather and climate

Bauer, Peter, Bjorn Stevens, and Wilco Hazeleger. "A digital twin of Earth for the green transition." *Nature Climate Change* 11.2 (2021): 80-83.

Schmude, Johannes, et al. "Prithvi WxC: Foundation Model for Weather and Climate." *arXiv preprint arXiv:2409.13598* (2024).

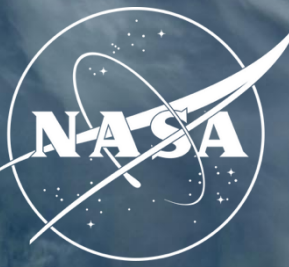


Data Assimilation and Weather Forecasting Pipeline



A reanalysis is created by taking historical observations and re-running them through a fixed, modern data assimilation system over a long time period (decades).

Reanalysis Comparison



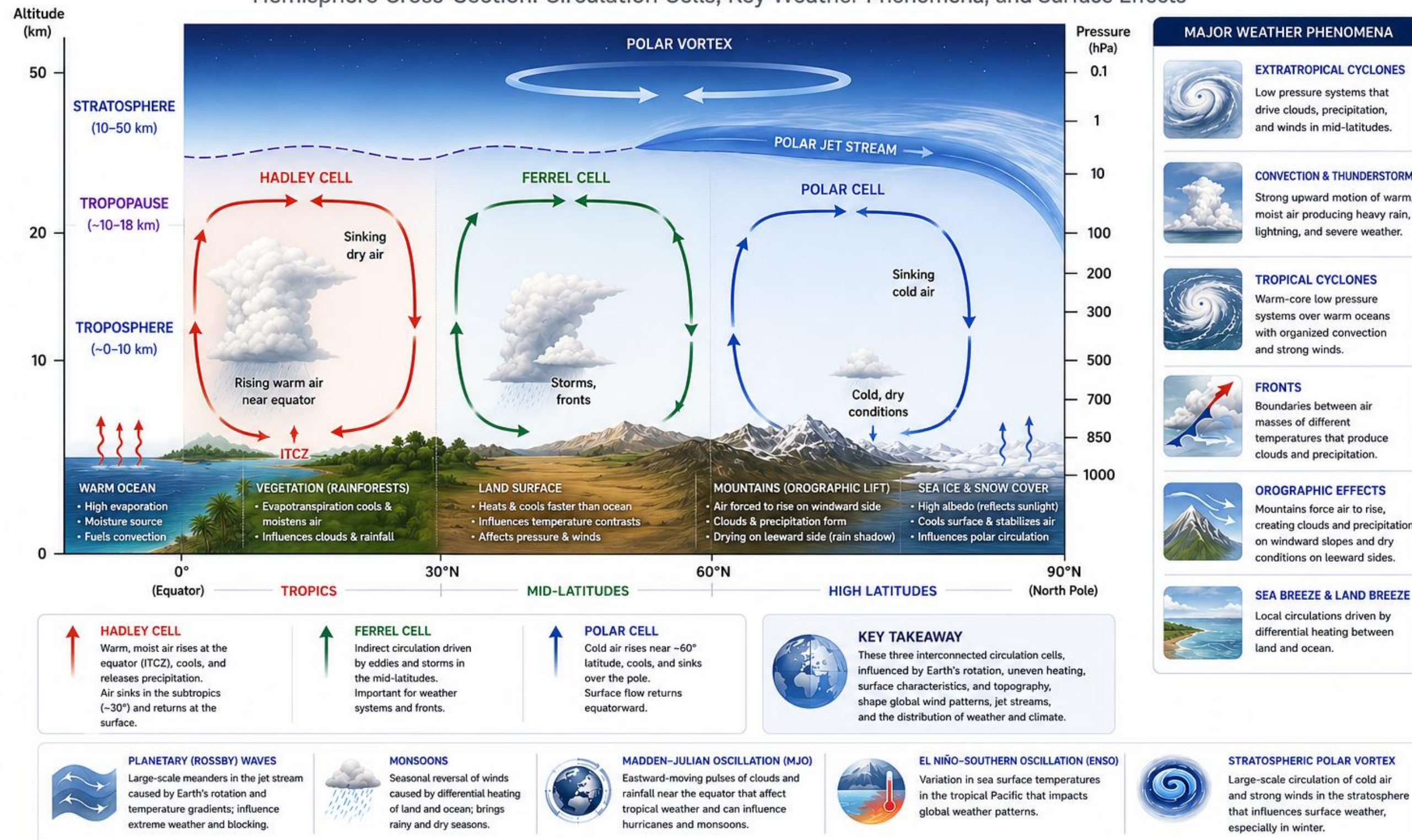
	ERA5	MERRA-2
Produced by	ECMWF	NASA GMAO
Full name	Fifth generation ECMWF reanalysis	Modern-Era Retrospective Analysis for Research and Applications version 2
Spatial resolution	0.25 degree	0.5 by 0.625 degree
Temporal resolution	Hourly	Hourly
Vertical levels	137 model levels	72 model levels
Data assimilation	4D Var	3D Var with incremental analysis update
Treatment below surface	Interpolates variables below surface	Uses missing values below surface
Source	- Single-level/surface: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview - Pressure level: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-pressure-levels?tab=overview	https://disc.gsfc.nasa.gov/datasets?project=MERRA-2 (all collections)
Documentation	https://confluence.ecmwf.int/display/CKB/ERA5%3A+data+documentation	https://gmao.gsfc.nasa.gov/media/publications/zbly36ziNFDfBmYmvhQeVqPhUo/Bosilovich785.pdf

Atmospheric Dynamics and Variability on Medium-Range Timescales



GLOBAL ATMOSPHERIC DYNAMICS

Hemisphere Cross-Section: Circulation Cells, Key Weather Phenomena, and Surface Effects



Latitudinal regions, dynamics, and area-weighting

Tropics (30°S–30°N)

- Large-scale, low-gradient flow dominated by convection and waves
- Smaller contribution to global RMSE, but critical for energy and moisture transport

Midlatitudes (30–60°)

- Strong gradients, baroclinic instability, storm tracks
- Dominates error growth and provides the most physically meaningful skill assessment

High latitudes (60–90°)

- Low variability, weak gradients, stable conditions
- Overrepresented without area-weighting and can artificially inflate skill

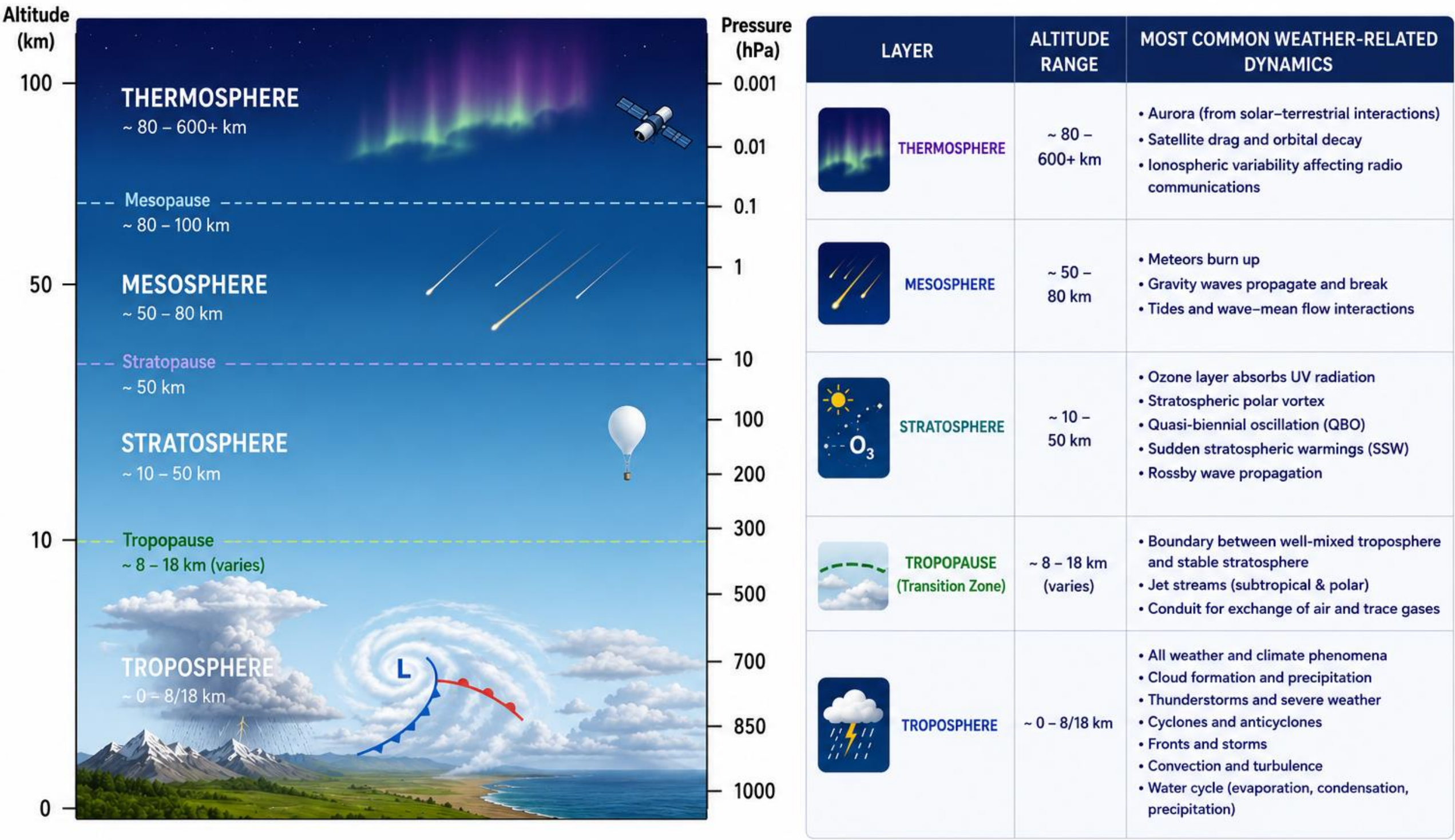
Area-weighting ensures each region contributes according to its true physical size, preventing high-latitude bias and preserving the dynamical relevance of tropical and midlatitude processes.

From the tropics to the poles, these interconnected systems operate across a wide range of scales, from small thunderstorms to planetary waves, shaping Earth's weather and climate.

Atmospheric Stratification



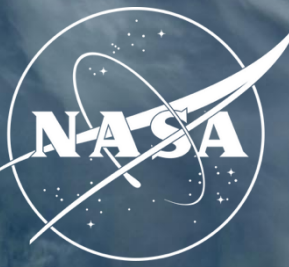
ATMOSPHERIC STRUCTURE AND WEATHER-RELATED DYNAMICS



Our plotting example code plots 3D variables at:

- 950 hPa: Surface (just above land)
- 850 hPa: Lower Troposphere
- 500 hPa: Upper Troposphere
- 200 hPa: Mid-stratosphere

Medium-Range Weather Forecasts



Medium-Range Forecasts (Days to ~2 Weeks)

- Driven mainly by atmospheric initial conditions
- Capture short-term weather systems (storms, fronts, cyclones)
- Skill declines quickly after ~5 days due to chaotic atmosphere
- Useful for operational weather prediction and near-term hazard response

GEOS-FP

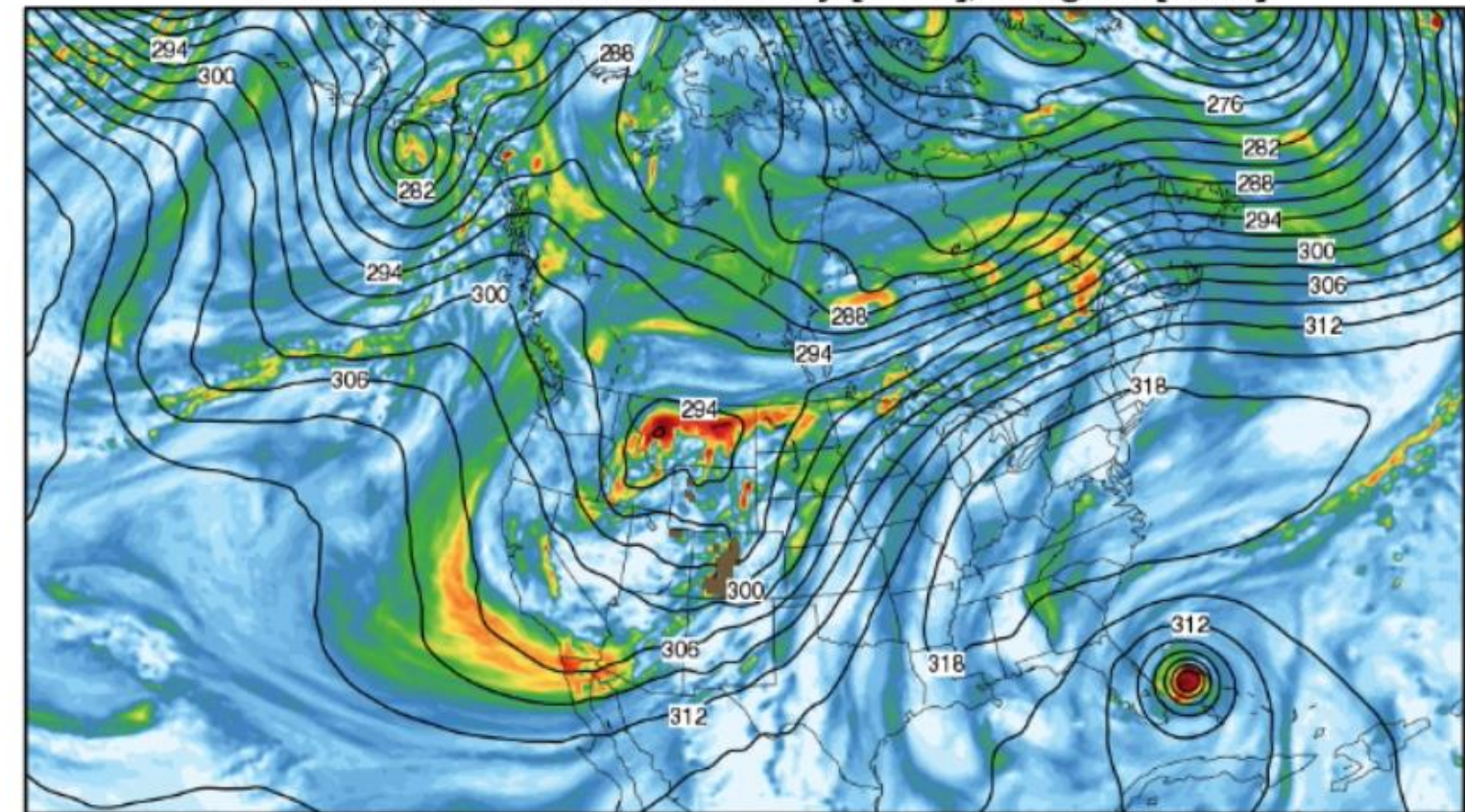
Weather Maps



GMAO

NASA/GMAO - GEOS Forecast Initialized on 00z 11/10/2022

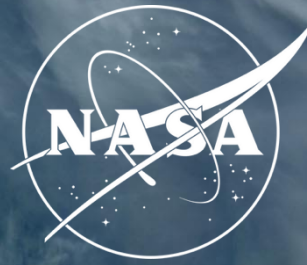
700 hPa Abs Potential Vorticity [PVU], Heights [dam]



Stop < > Refresh Rock Slower Faster Capture

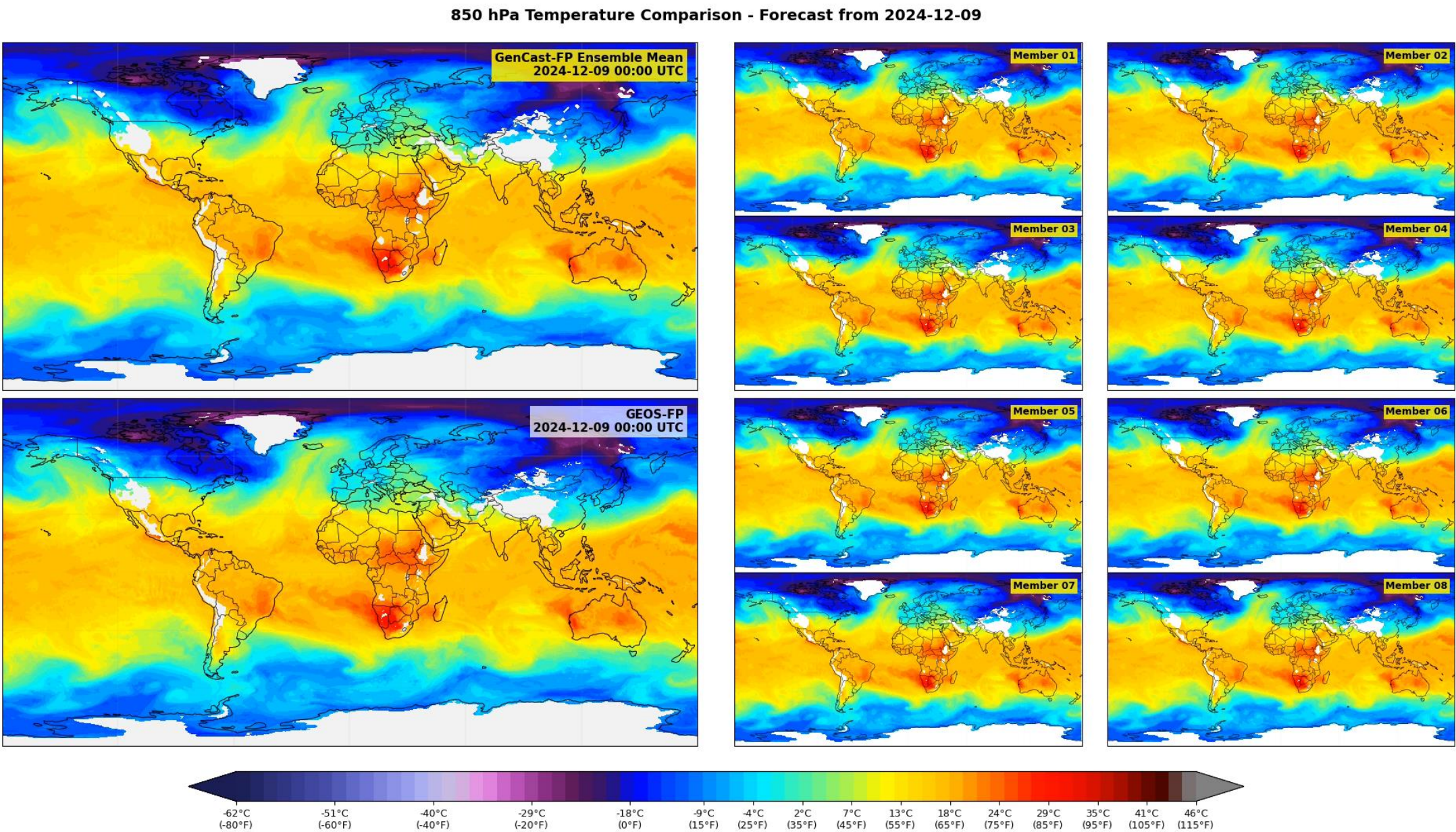
Stop < > Refresh Rock Slower Faster Capture

GEOS-AI Weather Prediction

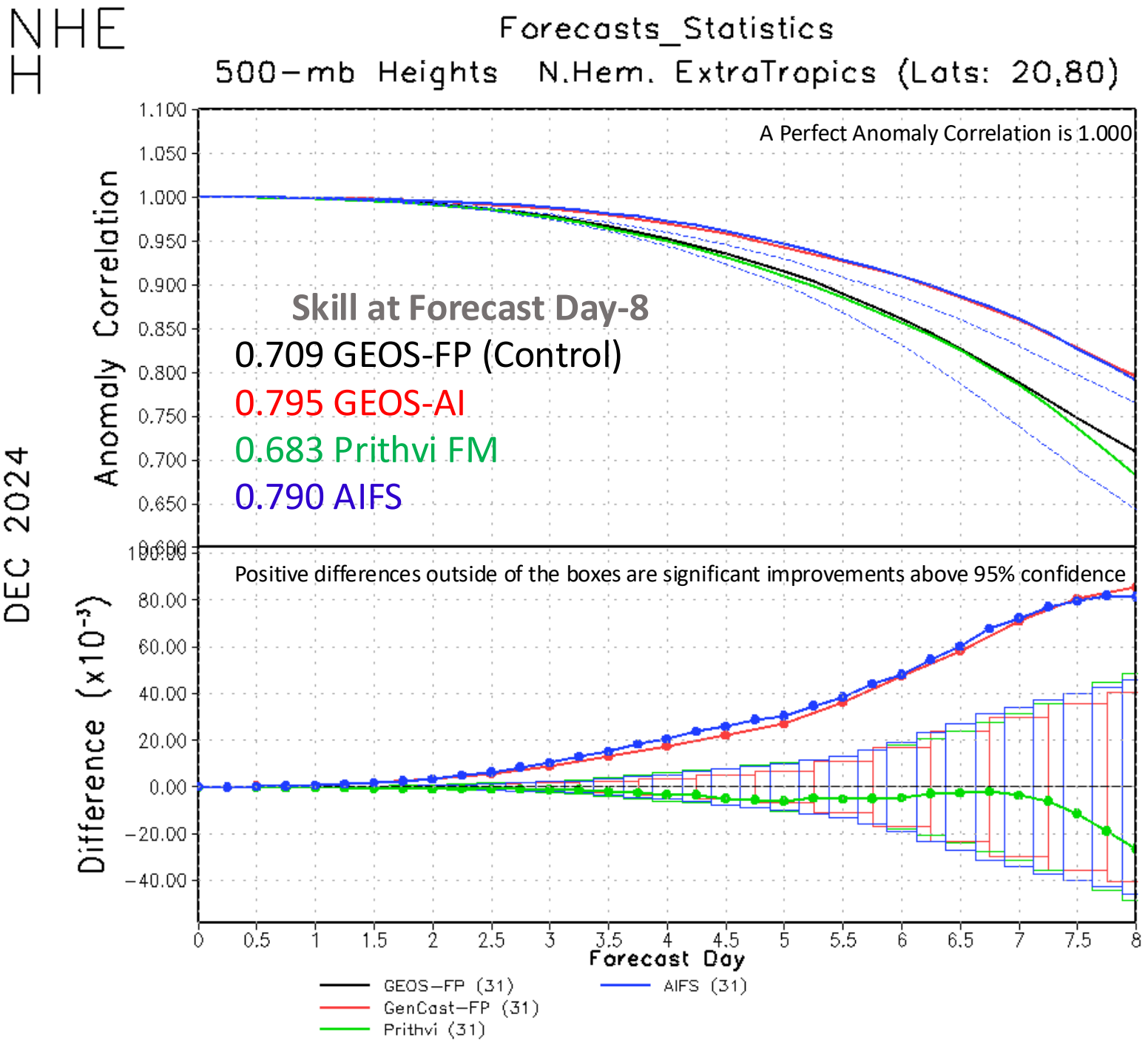


NASA GMAO Foundation Model Application for Weather Prediction

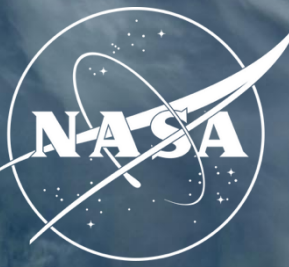
- **Evaluation:** NASA GMAO and the Goddard Data Science Group quantitatively evaluated foundation models (FMs) pre-trained on large-scale reanalysis datasets
- **Implementation:** GEOS-AI deploys the GenCast foundation model in real-time using GEOS-FP analyses as initial conditions
- **GEOS-AI Performance:**
 - Produces 10-day forecasts in minutes on a single GPU at NCCS
 - Generates ensemble predictions for atmospheric and surface variables
 - Significantly outperforms the physically-based GEOS-FP model
 - Global weather prediction skill comparable to ECMWF's leading AI model (AIFS)



NHE
H



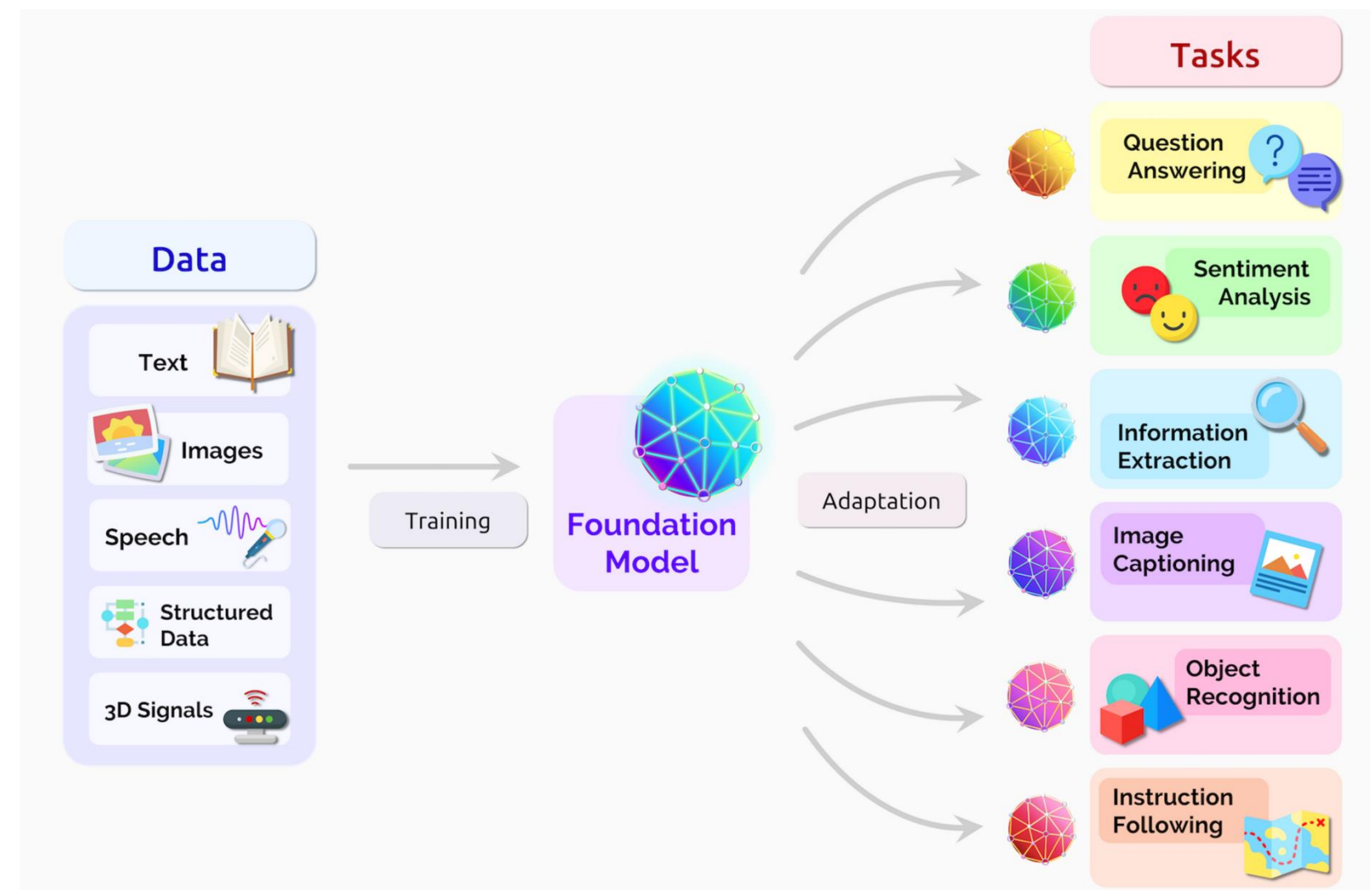
Background: Foundation Models



- The seminal Stanford white paper on foundational modeling introduced the concept of using large, pre-trained models to initialize a variety of downstream tasks

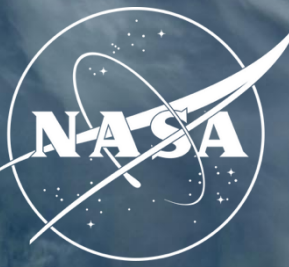
A foundation model can centralize the information from all the data from various modalities. This one model can then be adapted to a wide range of downstream tasks.

Seminal Stanford CRFM white paper, 2021



Bommasani, Rishi, et al. "On the opportunities and risks of foundation models." *arXiv preprint arXiv:2108.07258* (2021).

Foundation Models for Atmospheric Science



Goal

Quantitatively evaluate when, how, and under what conditions foundation models add scientific value to medium-range weather forecasts.

Conservation
Laws

Multivariate

3D time-
evolving fields

Structured/gridded
data

Train on reanalysis

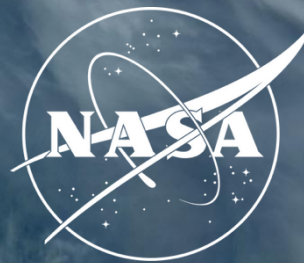
**Foundation Model
WxC**

***Physics/scale-aware**

*Constraints learned from reanalysis:

- Physics-based
 - solves atmospheric equations +/- parameterizations
- Multiscale
 - resolves large-scale flow
 - represents subgrid processes
- Data-assimilated
 - physically consistent in time with observations
- Approximately conservative
 - mass/energy approximately preserved
- Physically consistent reference for evaluation

Foundation Models for WxC Encoded Variables



Physical quantity	ERA5 variable name	Units	Vertical structure	Notes
Zonal wind (U)	u	m s^{-1}	$\approx 1000\text{--}50 \text{ hPa}^*$	East–west wind
Meridional wind (V)	v	m s^{-1}	$\approx 1000\text{--}50 \text{ hPa}$	North–south wind
Vertical velocity (W)	w	m s^{-1}	$\approx 1000\text{--}50 \text{ hPa}$	Geometric vertical velocity
Geopotential (Z)	z	$\text{m}^2 \text{ s}^{-2}$	$\approx 1000\text{--}50 \text{ hPa}$	Divide by g for height (m)
Specific humidity (Q)	q	kg kg^{-1}	$\approx 1000\text{--}50 \text{ hPa}$	Water vapor mass fraction
Temperature (T)	t	K	$\approx 1000\text{--}50 \text{ hPa}$	Air temperature
Mean sea level pressure (SLP)	msl	Pa	Surface	Standard sea-level pressure
2-meter temperature	t2m	K	Surface	Near-surface air temperature
10-m zonal wind	u10	m s^{-1}	Surface	Near-surface wind
10-m meridional wind	v10	m s^{-1}	Surface	Near-surface wind

Atmospheric state at time t

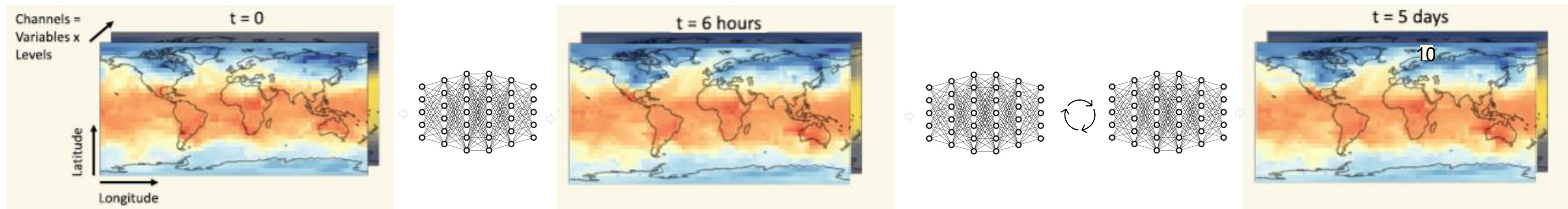


Foundation Model



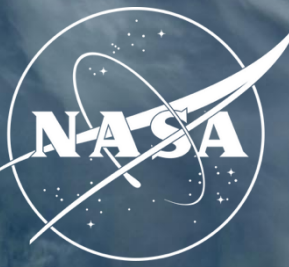
Atmospheric state at time $t + \delta$

*1000, 925, 850, 700, 600, 500, 400, 300, 250, 200, 150, 100, 50 hPa



Model	Dataset	NN Base Architecture	Resolution	Surface Vars	Atmospheric Vars	Num levels
Aurora	ERA5	Transformer	0.25° x 0.25°	2T, U10, V10, MSL	U, V, T, Q, Z	13
Pangu	ERA5	Transformer	0.25° x 0.25°	2T, U10, V10, MSL	U, V, T, Q, Z	13
GenCast	ERA5	GNN+Diffusion	1.0° x 1.0°	2T, U10, V10, MSL	U, V, w, T, Q, Z	13
AIFS	ERA5	GNN	0.25° x 0.25°	2T, 2DT, U10, V10, MSL, SP, TS	U, V, w, T, Q, Z	13
Prithvi-WxC	MERRA2	Transformer	0.5° x 0.625°	2T, U10, V10, MSL, SP, TS, etc. (20)	U, V, w, T, Q, Z, etc. (10)	14

Key differences between SOTA FMs: Vision Transformer (ViT) vs Graph Neural Network (GNN)



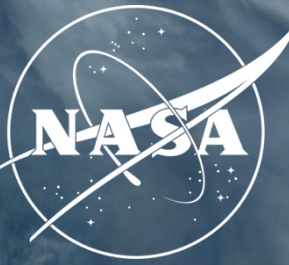
Transformers (ViTs, CNN-based)

- Treat data as a grid or sequence of tokens
- Use self-attention to capture global relationships
- Strong at modeling long-range dependencies
- Typically operate on regular grids (lat–lon)
- Scale well with data, but computationally expensive ($O(N^2)$)

Graph Neural Networks (GNNs)

- Represent data as a graph (nodes + edges)
- Encode spatial relationships explicitly (e.g., Earth geometry)
- Naturally handle irregular grids / meshes
- Use message passing for local-to-global information flow
- Often more efficient and physically structured, very computationally heavy

NN Architecture Implications



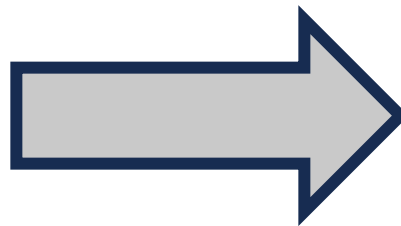
Graph-based models
(GenCast, AIFS)



Better anomaly capture

Global message passing inherent in GNNs accelerates information/error propagation and smooths unresolved scales

**Transformer (e.g. CNN)-
based models**
(Aurora, Pangu)



Preserve local spatial consistency, resulting in slower amplitude degradation

More tightly coupled to reanalysis used for training

Models differ in:

- Variables
- Resolution
- Training periods
- Preprocessing
- Architectures
- etc...

==> Reported skill is not a fair comparison unless compared to a standard/benchmark

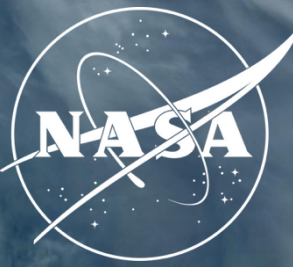
What is a benchmark?

- Standardized dataset
- Fixed evaluation protocol
- Agreed metrics

Purpose:

- Fair comparison
- Reproducibility
- Track progress and development

Core benchmarks for medium-range forecasts



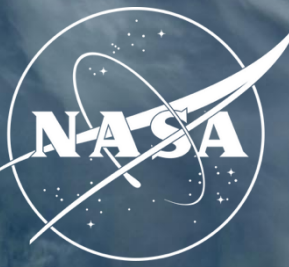
Benchmark	Fields and Range	Typical Use	Reference
WeatherBench	Global geopotential, temperature, precipitation; 3-5 days	Early benchmark for ML vs NWP	Rasp, Stephan, et al. "WeatherBench: a benchmark data set for data-driven weather forecasting." Journal of Advances in Modeling Earth Systems 12.11 (2020): e2020MS002203.
WeatherBench2	Global geopotential, temperature, precipitation; 1-14 days	Next-gen; SOTA models vs operational centers	Rasp, Stephan, et al. "Weatherbench 2: A benchmark for the next generation of data-driven global weather models." Journal of Advances in Modeling Earth Systems 16.6 (2024): e2023MS004019.

Shared components:

- ERA5-based datasets
- Standard splits (train/val/test)
- Metrics:
 - RMSE
 - ACC
- Baselines:
 - Climatology
 - Persistence
 - NWP (e.g., IFS)

Note: these benchmarks are for use with deterministic models – WeatherBench Probability is available for probabilistic and ensemble benchmarking

Reanalysis as a benchmark



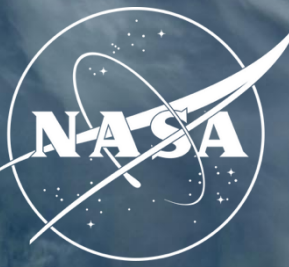
	WeatherBench	This Session (ERA5)
Data source	ERA5 (preprocessed)	ERA5 (raw/controlled subset)
Standardization	High	Moderate
Flexibility	Low	High
Reproducibility across papers	High	Lower
Real-world workflow	Lower	Higher
Out-of-sample control	Limited	Explicit

Bottom line: for research papers, WeatherBench is useful, but too limited for detailed evaluation

Forecast Evaluation

Benchmarking approaches for weather and climate
Overview of forecast skill metrics

Evaluation Philosophy

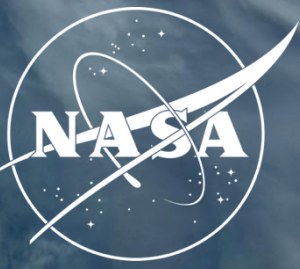


- We are evaluating forecast skill relative to a physically consistent reference (reanalysis, NWP)
- We compare absolute error (RMSE/difference) and pattern skill (ACC)
- We assess regional dynamics (global vs tropics vs hemispheres)
- We assess vertical structure (pressure levels)
- We assess forecast degradation with lead time

Key message:

No single metric tells the story. Skill depends on region, variable, scale, and lead time.

Metric Description: Bias



$$\text{Bias} = \frac{1}{N} \sum_{i=1}^N (f_i - o_i)$$

f is the forecast
 o is the reference

Difference (bias / error maps)

Shows systematic spatial structure of errors

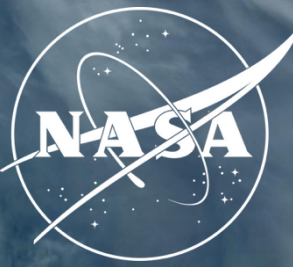
Best for diagnosing:

- regional biases
- regime-dependent failures
- land-ocean contrast issues
- storm track shifts

Interpretation:

- Persistent patterns = model bias
- Random/noisy = phase error / chaotic growth

Metric Description: Anomaly Correlation Coefficient (ACC)



$$ACC = \frac{\sum_{i=1}^N f'_i a'_i}{\sqrt{\sum_{i=1}^N (f'_i)^2 \sum_{i=1}^N (a'_i)^2}}$$

where:

f'_i is the forecast anomaly (forecast-climatology)

$$f'_i = f_i - \overline{clim_i}$$

a'_i is the analysis anomaly (analysis - climatology)

$$a'_i = a_i - \overline{clim_i}$$

ACC = 1 ---> perfect pattern match

ACC = 0 ---> no forecast skill

Measures pattern similarity, not magnitude
Removes climatology → focuses on anomalies

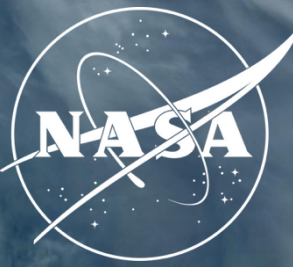
Interpretation:

- High ACC = correct large-scale structure
- Can look artificially good when:
 - variance is low (e.g., poles)
 - fields are smooth

Critical point:

A model can have high ACC but still be physically wrong in amplitude (RMSE catches this)

Metric Description: Root Mean Squared Error (RMSE)



Measures magnitude of error → Grows with increasing lead time (time from initial conditions)

Sensitive to:

- gradients (midlatitudes)
- extremes

Interpretation:

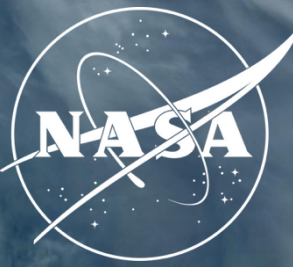
- Fast growth → poor predictability handling

Compare across regions:

- Midlatitudes → highest RMSE growth (baroclinic chaos)
- Tropics → lower RMSE but can hide structural issues

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (f_i - a_i)^2}$$

Area-Weighted Forecast Comparison Metrics



```
## Area Weighted Metrics:
weights = np.cos(np.deg2rad(lat))
weights = weights / weights.mean()

rmse = np.sqrt(((f - a)**2 *
weights).mean(dim=["lat", "lon"]))

acc = (
    ((f_anom * a_anom) *
weights).sum(dim=["lat", "lon"]) /
    np.sqrt(
        ((f_anom**2 * weights).sum(dim=["lat", "lon"]))
    ) *
    ((a_anom**2 * weights).sum(dim=["lat", "lon"]))
    )
)
```

Why area-weighted metrics matter

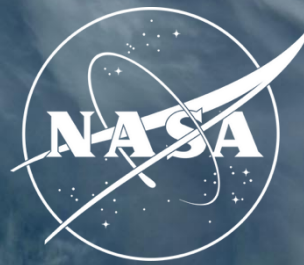
Earth system data are defined on a spherical grid, but latitude-longitude grids are not uniform in physical area.

Grid cells become progressively smaller toward the poles. Treating every grid point equally therefore gives disproportionate influence to high latitudes.

Area weighting corrects this by ensuring each location contributes in proportion to its true physical area.

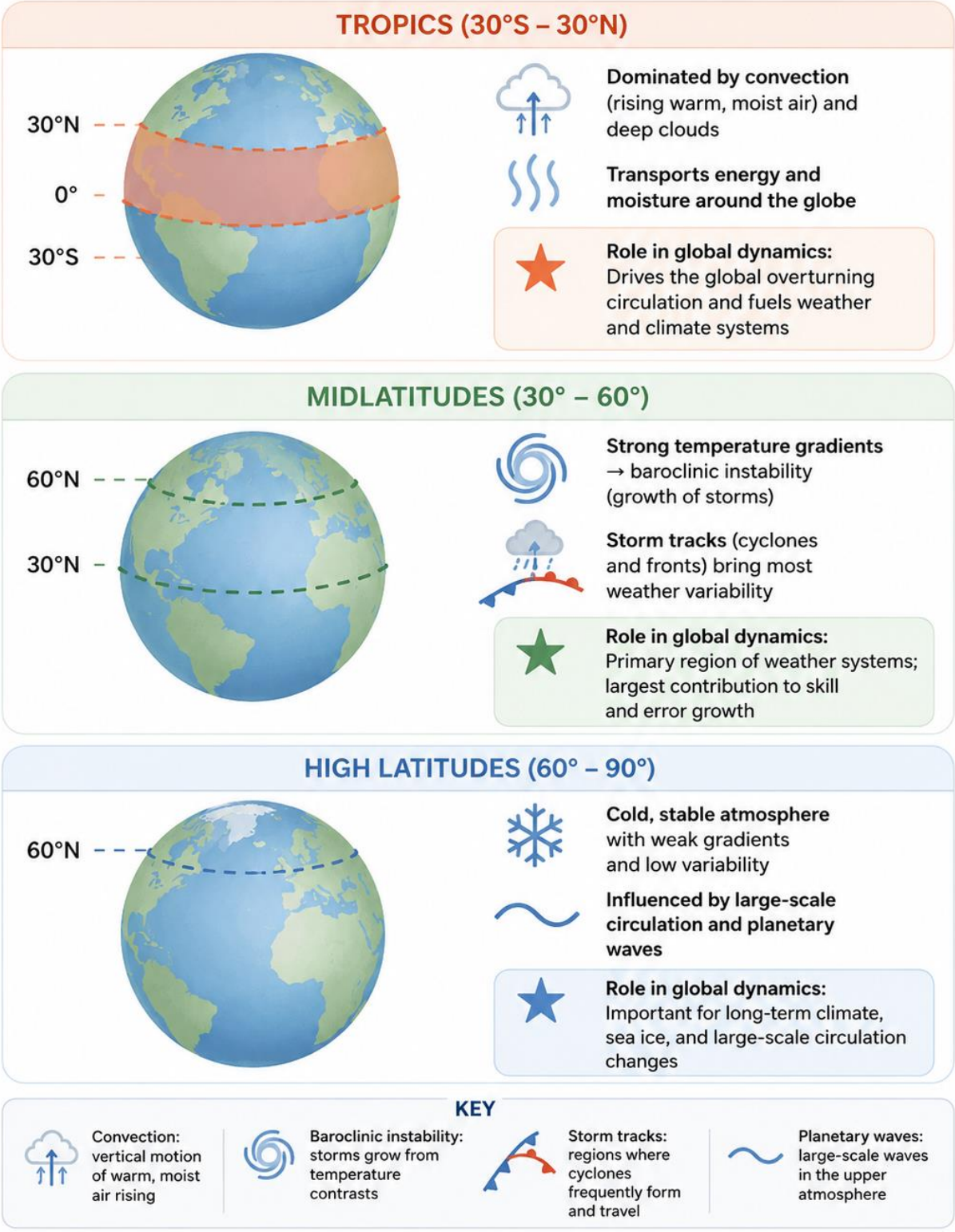
Area weighting is not a technical detail. It is required for physically meaningful, unbiased evaluation of global weather and climate forecasts.

Typical Metric Comparisons for Robust Forecasts

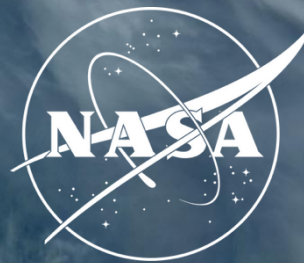


Region	Signal characteristics	Metric behavior	Risk
Global	Mixed regimes; tropics dominate area, midlatitudes dominate variance	High ACC (tropical influence), RMSE dominated by midlatitudes	Masks regional differences; misleading summary
Tropics	Large-scale, low amplitude	High ACC, low RMSE	Overestimating skill
Midlatitudes	High gradients, chaotic dynamics	Lower ACC, high RMSE growth	Most honest comparison
Poles	Low variance, weak gradients	High ACC, low RMSE	Artificially good scores

LATITUDINAL BANDS: Dynamics and Their Role

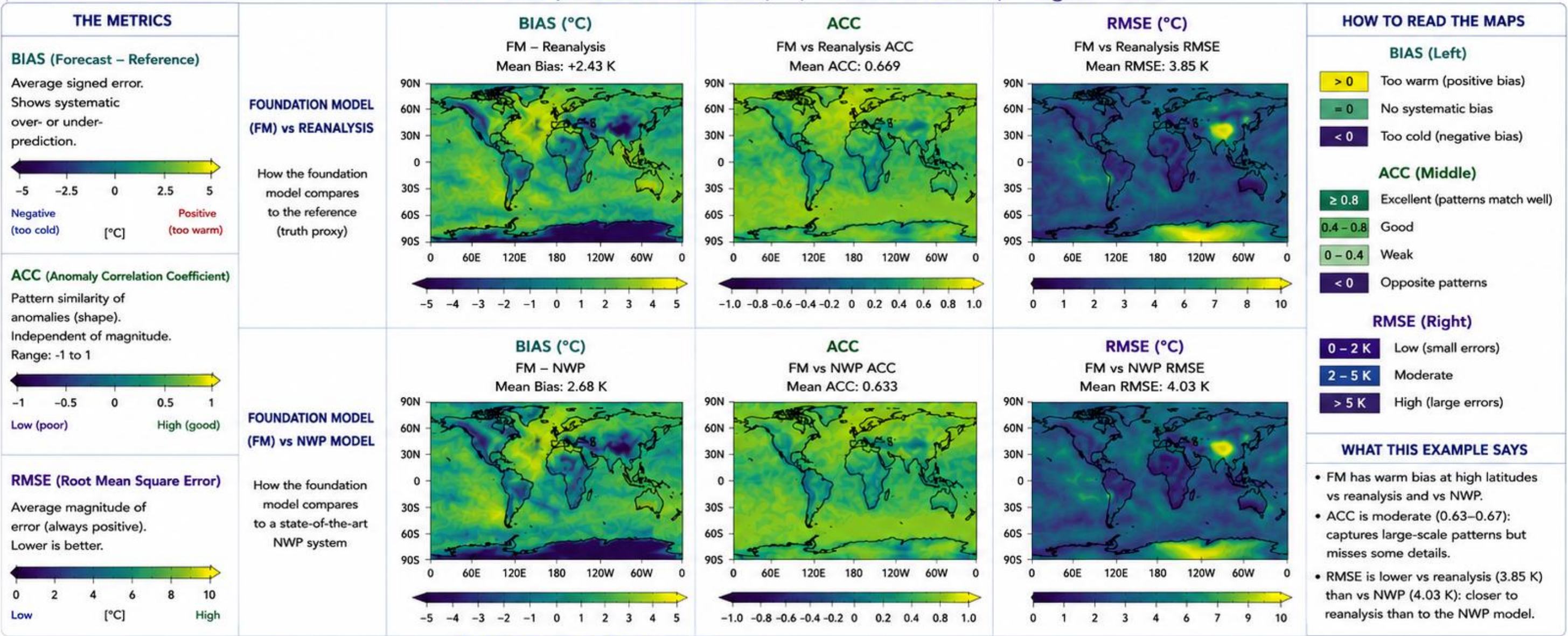


Interpreting Forecast Metrics



HOW TO INTERPRET: BIAS, ACC, AND RMSE MAPS

Example: 500 hPa Temperature at Forecast Hour 120
Left = Bias (Forecast – Reference) | Middle = ACC | Right = RMSE



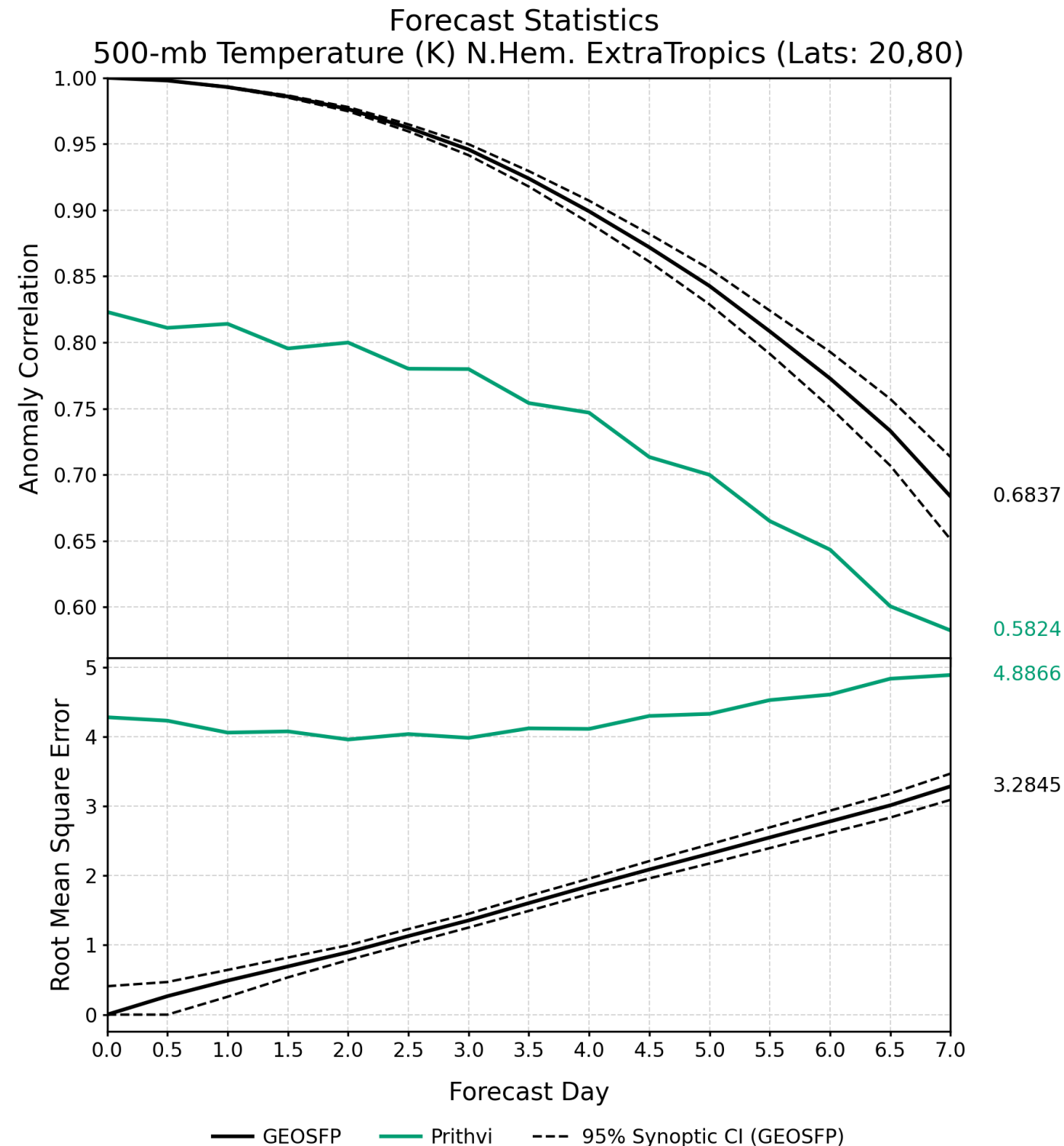
Example: Prithvi WxC 500 hPa Temperature (K) at 120 hour (5 day) lead time

What to look at first (at 120 hours)

At ~Day 5:
ACC → pattern skill (is the structure right?)
RMSE → magnitude error (is the amplitude right?)

This is the decision point where models start to separate meaningfully





120 hour lead time metrics:

ACC

NWP (black): ~0.68

FM (green): ~0.58

RMSE

NWP: ~3.3 K

FM: ~4.9 K

Interpretation

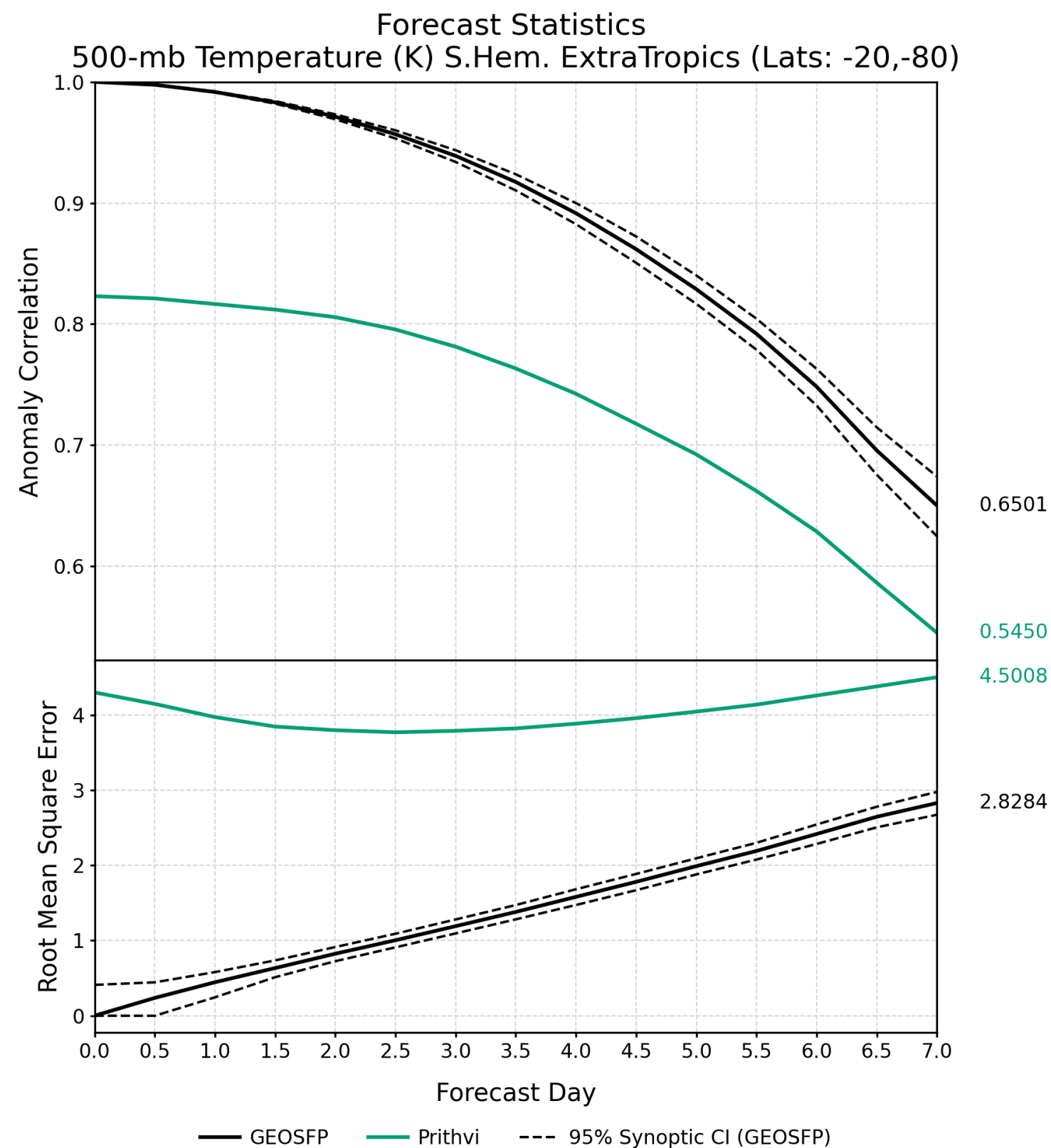
- FM has lower pattern skill and larger errors
- This is the hardest regime (baroclinic storms, strong gradients)

What this likely means physically

- Storm tracks displaced or smeared
- Wave phase errors (troughs/ridges shifted)
- Possibly weaker gradients → under-amplified systems

Key takeaway

In NH midlatitudes, the FM is not maintaining synoptic structure as well as NWP



120 hour lead time metrics:

ACC

NWP: ~0.65

FM: ~0.55

RMSE

NWP: ~2.8 K

FM: ~4.5 K

Interpretation

- Same pattern as NH, but slightly cleaner signal
- FM again shows:
 - **lower ACC**
 - **higher RMSE**

Physical meaning

Jet and storm track errors

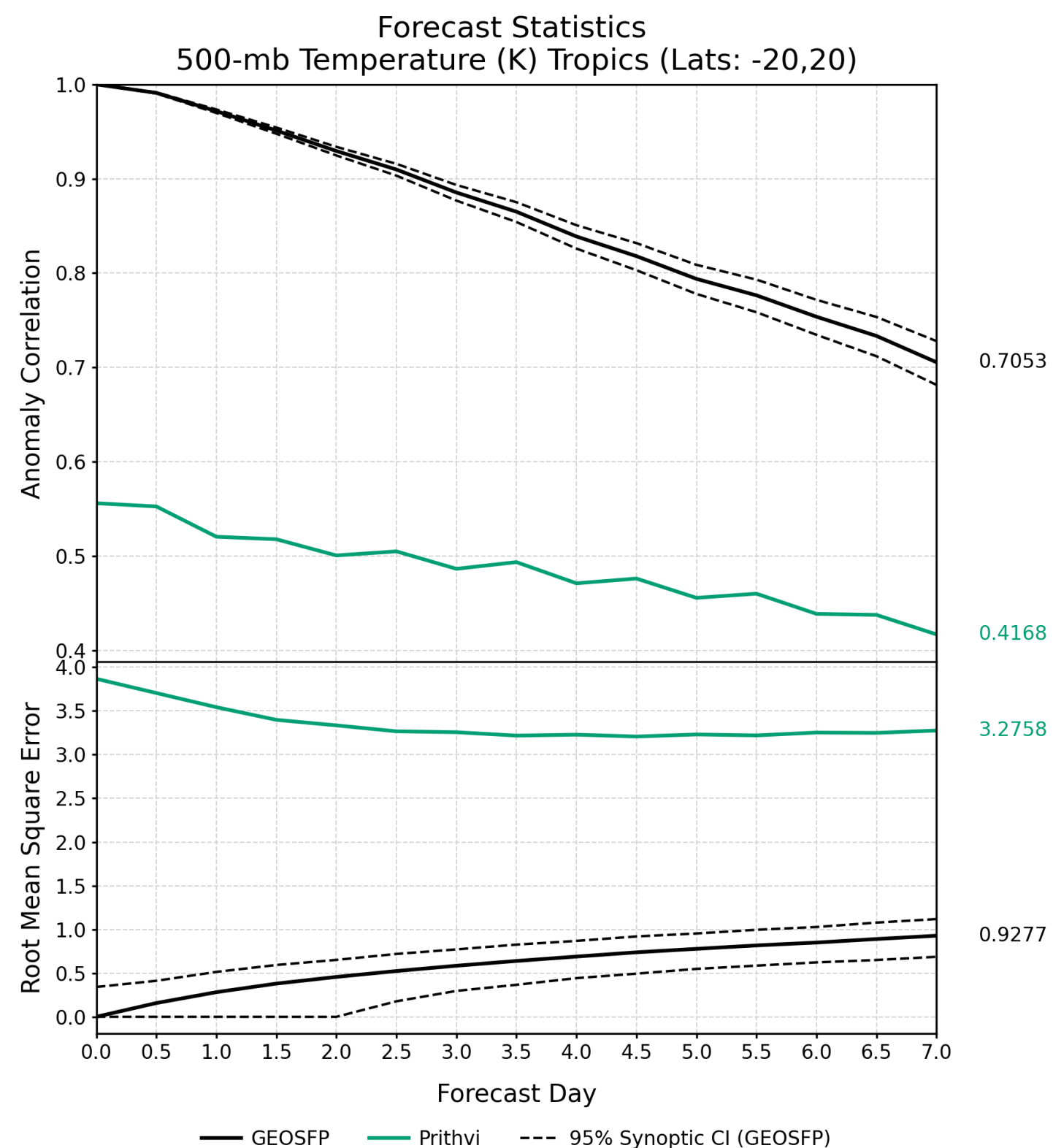
- Likely:
 - zonal smoothing
 - phase lag in waves

Subtle point

- SH is more zonally symmetric → errors often look like:
 - shifted belts, not localized failures

Key takeaway

FM struggles with large-scale wave structure even in simpler, more zonal flow



120 hour forecast metrics:

ACC

NWP: ~0.70

FM: ~0.42 (!!)

RMSE

NWP: ~0.9 K

FM: ~3.3 K

Interpretation

Largest degradation for FM

Both:

very low ACC

high RMSE

Physical meaning

This is not just amplitude error, it's structural:

Convection in wrong place (ITCZ shifts)

Poor wave representation (Kelvin, MJO, etc.)

Tropics often have:

lower RMSE sensitivity

BUT here RMSE is still large → serious mismatch

Key takeaway

FM is not capturing tropical organization or variability correctly at Day 5

**THANK
YOU**

ATTENTION

FOR YOUR